CLAIMS

What is claimed is:

1. A computer system, comprising:

- a processor having a first massively parallel optical interface;
- a motherboard having a processor socket configured to optically couple with the processor's first optical interface, the motherboard further comprising at least one optical memory slot; and
- at least one memory module having a second massively parallel optical interface, said memory module being installed in the optical memory slot to create an optical data pathway between the processor and the memory module.

2. The system of claim 1, wherein the first and second optical interfaces each comprise a two-dimensional array of photon emitters or photon sensors.

3. The system of claim 1, wherein the data is transmitted between components as a sequence of two-dimensional frames of light.

4. The system of claim 1, wherein the motherboard further comprises at least one optical peripheral port for connection to an external device via an optical cable.

5. The system of claim 1, wherein the aggregate data bandwidth of the system's internal optical data pathways is in the petabit-per-second range.

6. The system of claim 1, wherein the optical data pathway between the processor and the memory module eliminates the need for a traditional electrical memory bus.

7. A method of operating a computer system, comprising the steps of:

- transmitting data from a processor to a motherboard via a first massively parallel optical interface;
- routing said data across optical pathways on the motherboard; and
- transmitting said data from the motherboard to a memory module via a second massively parallel optical interface.

8. The method of claim 7, wherein the data is transmitted as a sequence of frames, with each frame containing millions of parallel data channels.

9. A three-dimensional integrated circuit, comprising:

- a first semiconductor die containing a first circuit;
- a second semiconductor die physically stacked on top of the first die, the second die containing a second circuit;
- a first array of photon emitters or sensors on a surface of the first die; and
- a second array of photon emitters or sensors on a corresponding surface of the second die, positioned to create a massively parallel optical data bus for inter-layer communication between the first and second circuits.

10. The integrated circuit of claim 9, wherein the optical data bus provides a data transfer rate in the terabit-per-second or petabit-per-second range between the dies.